

Foldable Long-Distance Ball Transfer Mechanism - Midterm Design Presentation

This English version is formatted for graduate application review. It preserves the engineering purpose, design decisions, validation evidence, and individual contribution signals of the original course document while presenting the material in concise English.

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Project	Foldable Long-Distance Ball Transfer Mechanism
Course	Machine Design Theory
Document Type	Midterm design presentation
Primary Role	Team lead and concept designer
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Executive Summary

The midterm presentation documented concept-stage work for the foldable mechanism, including container design, polar-positioning arm logic, packaging constraints, and early architecture decisions.

The key contribution was identifying that packaging and deployment reliability would dominate the final design. This helped the team avoid building a mechanism that worked in theory but could not fit, lock, or survive testing.

Applicant Contribution Signals

- Defined packaging constraints as primary design inputs.
- Compared early architecture options around motion, size, stiffness, and fabrication.
- Prepared the project for later simplification and reliable hardware choices.

Concept Direction

The project explored how a ball-transfer mechanism could be stored compactly while still deploying into a useful working geometry.

Early work considered container layout, polar-positioning arm behavior, and the mechanical consequences of strict size limits.

Path to Final Design

The midterm stage made clear that the final mechanism would need simpler hardware, predictable deployment, and careful assembly planning.

Technical Decision Table

Constraint	Decision	Why It Mattered	Skill Demonstrated
Packaging constraint	Centered design around storage envelope	Made the main risk visible early	Requirement analysis
Motion concept	Evaluated polar-positioning arm logic	Connected geometry to mechanism function	Mechanism synthesis
Buildability	Prepared for accessible materials and hardware	Improved later final reliability	Fabrication planning

How to Read This Evidence

This document is intended to help a reviewer quickly understand the engineering content of the original report in English. The original PDF remains available in the project archive as the source artifact with its original figures, tables, screenshots, and course formatting.